

- I. **True/ False:** Read the following statements carefully and identify whether the statement is TRUE or FALSE. (10 * 0.5 marks = 5 marks)

1	The digital divide is a cyberethics concern	T
2	Whistle-blowing is considered a moral duty if the whistle-blower can prove their actions will prevent harm	* T
3	A Ponzi scheme and a pump-and-dump scam are the same thing	F
4	Floridi's Macroethics focuses only on human agents.	F
5	Biometric surveillance does not impact privacy because it is used for security purposes.	F
6	Computer-related careers typically require formal accreditation or apprenticeship	* F
7	Virtue Ethics focuses primarily on the consequences of actions	* F
8	Bahrain's IT Crimes law does not punish suspects who use encryption to conceal suspicious activities	F
9	Social Learning Theory emphasizes peer influence in shaping cybercriminal behaviour	T
10	Software piracy refers to unauthorized duplication and distribution of propriety software	T

II. **Multiple Choice:** Read each question carefully and select the one best answer by marking the corresponding letter. (10 * 0.5 = 5 marks)

1. According to Kant, actions have moral worth only when:
 - ☒ a. They are performed from duty
 - b. The benefit the majority
 - c. They achieve good results
 - d. They are culturally acceptable
 2. What does technological neutrality suggest?
 - a. Technology inevitable imposes its own behaviour
 - ☒ b. Technology is a neutral force and depends on human usage
 - c. Technology promotes ethical behaviour and becomes unethical because of human intervention
 - d. Technology is biased toward societal norms
 3. Which is a key requirement for whistle-blowing under De George's model?
 - a. The whistle-blower must be anonymous
 - ☒ b. The harm must be serious
 - c. The employee must publish evidence of corruption on social media
 - d. The whistle-blower must resign first
 4. What is a key feature of a fully developed profession?
 - a. The ability to conduct unlicensed work
 - ☒ b. Certification and accreditation process
 - c. Informal guidelines for conduct
 - d. No need for professional development
 5. Darknet markets are commonly associated with:
 - a. Legal software distribution
 - ☒ b. Illicit drug trafficking and human trafficking
 - c. Cryptocurrency trading
 - ~~d. All of the above~~
- Correct →
- 4.5

4. What are the three stages of money laundering, and how do cybercriminals use cryptocurrency at each stage? (4 marks)

- ① Placement → they will purchase cryptocurrency through the exchange like Peer to peer, or Crypto ATM.
- ② Layering → use unregulated exchange to trade coins into different currencies and use mixer.
- ③ Integration → the funds are introduced into legitimate and purchase the luxury goods or real estate.

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5. Compare the Privacy and Data Protection concepts based on Focus and Responsibility. Complete the table provided below. (4 marks)

Aspect	Privacy	Data Protection
Focus	Protection individual rights to and anonymity over personal information	Securing data to protect prevent breaches or unauthorized use
Responsibility	It will shared between individual or person and organization	It's primary organization and regulators that collect and process data

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III. Short Answer Questions: Please write clear, concise, and complete answers to the following questions (30 marks)

1. Explain the difference between morals and ethics. (2 marks)

Morals → ^{standards} ~~standards~~ and behavior will know by education. 1.5

ethics → ^{investigation} ~~collect~~ the information, and then analysis apply to ~~either~~ ^{ethical} theories to ~~individual or group~~ ^{investigation} to know right or wrong ^{ethical} law or medical

2. Why is the computing profession considered less developed than law or medical professionals? Use Ford and Gibbs's model of a fully developed profession to justify your answer (4 marks)

Because will get certifications, ^① ~~licensing~~ ^② skills development, ^③ society development, ^④ Accreditation, ^⑤ licensing 4

computing profession if they don't have Certificate in IT, they can work by talking courses or any certification, except law or medical because it's human life.

3. Describe the four components of Lessig's 'Pathetic Dot Theory', and how does it apply to cyberspace? (6 marks)

① Law → copyright and patient protection ~~regular~~ regulate behavior ~~and by prohibiting~~ certain activities.

② Market → Commercial and personal interaction in cyberspace in ~~the~~ various.

③ Code → programs and protocols can constrain and control our activities. Architecture

④ Norm → will ~~changing~~ set of Norms that regulate cyberspace include ~~network~~ social customs.

6. The primary reason for having a Code of Ethics is to:

- a. Prevent professionals from committing crimes
- b. Guarantee personal career success
- c. Increase the number of professionals in the field
- ☒ d. Ensure professional actions benefit society and minimize harm

7. Which term describes fraudulent cryptocurrency promotions designed to lure investors?

- a. Bounty hunting
 - ☒ b. ICO Fraud
 - c. Cryptojacking
 - ~~d. Exchange scam~~
- Correct →

8. What actions qualify as computer sabotage under Bahrain's IT Crimes law?

- a. Using someone's Wi-Fi without permission
- ☒ b. Installing ransomware on a company's network
- c. Copying a file from a public database
- d. Sending a personal email from a work computer

9. What is an example of secondary use of data?

- a. Encrypting user data for protection
 - ☒ b. A company selling customer data to advertisers without consent
 - c. Prevent all forms of data processing
 - ~~d. A government agency using data for national security~~
- Correct →

10. What type of cybercrime involves obtaining sensitive information by tricking users?

- ☒ a. Phishing
- b. Hacking
- c. Computer Sabotage
- d. Software piracy

IV. **Case Study Questions:** Please read each case study carefully and answer the following questions for each question (20 marks)

1. Amira, a junior software developer, works for a small health tech startup that just released a fitness-tracking app called FitPulse. The app monitors GPS location, heart rate, step count, and sleep patterns. During development, Amira raised concerns that the app lacks data encryption and stores user information in plain text on company servers. Her team lead tells her, "We'll address security in a future update—right now, our priority is launching and gaining users."

The company also shares user data with third-party advertising partners without informing users or requesting consent, arguing that it's mentioned in vague legal terms deep in the privacy policy. Amira feels uneasy but continues working, assuming the legal and marketing teams have approved everything.

Months after launch, a cybersecurity researcher discovers that FitPulse transmits real-time user data—including GPS coordinates and health metrics—over unencrypted HTTP connections. Worse, a group of attackers exploited this flaw to track user movements and habits, including that of a high-profile public figure. The news sparks public outcry. Amira, troubled by how the company handled the situation, begins questioning her role and responsibilities.

- a) By utilizing the ACM Code of Ethics and Professional Conduct and the IEEE Code of Ethics in the last page of this exam, determine which ethical principles are relevant to this scenario? (8 marks)
- b) What should a junior developer do when ethical concerns are dismissed by leadership? (4 marks)

2. Rashid was a senior system administrator at GulfXpress, a mid-sized logistics company that handles sensitive shipment data for government contracts and international clients. After a series of disagreements with his manager over policy changes and workload, Rashid was terminated without prior warning.

Although HR initiated the process to revoke his access, a miscommunication led to his admin credentials remaining active for 48 hours after his dismissal. During that time, from his home network, Rashid logged into the company's internal system and deleted logistics reports and operational logs related to upcoming shipments, encrypted the customer database using a ransomware script he had written himself months earlier, left a note on the system dashboard, demanding BD 1,000 in cryptocurrency to unlock the encrypted data and threatened to leak customer shipping details if the company did not comply within 72 hours.

IT staff discovered the intrusion during their morning shift. After a digital forensic analysis, the breach was traced back to Rashid's IP address. The company involved local authorities and filed a formal complaint under the Information Technology Crimes Law.

- a) Under the Bahraini Information Technology Crimes law, which articles did Rashid violate through his attacks on GulfXpress? (4 marks)
- b) As a cybersecurity expert, what technical and procedural safeguards could GulfXpress have implemented to prevent these attacks? (4 marks)

① Ransomware : (up to 3 years) , 30,000 BD

electronic-break-in (no more 1 year), 30,000 BD

encryption : (up to 3 years) , 100,000 BD

② - first Backup everyday to don't deleted all data

- Awareness to employee ~~to don't active in any~~ in cyber rules.

- don't give all data or sensitive to any new employee.

Best of Luck!